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Limited added value of recall-derived dietary variables for cardiovascular disease classification in NHANES, 2007–2018

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Abstract

Short-term recall-derived dietary variables may be relevant to cardiovascular disease, but their classification value after demographic, lifestyle, and clinical markers are observed is unclear in cross-sectional survey data. We analyzed pooled data from 34 388 United States adults in the National Health and Nutrition Examination Survey, 2007–2018, after excluding participants with missing outcome status or current pregnancy. The outcome was a self-reported cardiovascular disease composite defined from coronary heart disease, angina, myocardial infarction, or stroke. We evaluated 3 nested predictor blocks: demographic and lifestyle variables, clinical and laboratory variables, and recall-derived dietary variables. Logistic regression and histogram-based gradient boosting were assessed with 5-fold out-of-fold validation. Adding the clinical block produced the largest discrimination gain (Δ area under the receiver operating characteristic curve 0.031 for logistic regression and 0.032 for gradient boosting; both $P < .001$). Adding dietary variables after the clinical block produced a 0.002 gain in logistic regression and no material gain in gradient boosting ($P = .219$). Weighted logistic regression, subgroup, and repeated stochastic imputation sensitivity analyses preserved the same hierarchy. In this setting, recall-derived dietary variables added little classification information once clinical markers were available.

Key words cardiovascular disease, dietary recall, cross-sectional classification, incremental discrimination, NHANES, prediction modeling, epidemiologic methods

Introduction

Cardiovascular disease remains a leading source of morbidity and mortality worldwide and in the United States, and scalable tools for surveillance and cross-sectional phenotype classification remain of interest.[1-3] Most cardiovascular prediction work, however, has focused on incident-risk equations, recalibration, and external validation rather than on the order in which common predictor domains add classification information in cross-sectional survey data.[4-7] Some studies have also linked dietary behavior to predicted 10-year cardiovascular risk through existing equations, which does not answer the residual-information question once downstream clinical markers are already observed.[8]

Studies using the National Health and Nutrition Examination Survey (NHANES) often combine demographic, behavioral, clinical, and dietary variables in a single feature set and then report overall model performance or variable rankings.[9-12] That strategy is useful for benchmarking, and related cardiovascular machine-learning work spans meta-analytic syntheses, coronary calcium models, and biobank-based actionable risk equations.[13-16] But it does not isolate a narrower epidemiologic question: after demographic, lifestyle, and clinical information is available, how much additional cross-sectional classification signal is carried by recall-derived dietary variables?

That question is practically relevant because a one- or two-day 24-hour recall reflects short-term intake rather than habitual diet, and nutritional epidemiology has long emphasized cautious interpretation of such measurements, including in relation to total energy intake and usual intake.[17-19] Dietary exposures remain biologically relevant to cardiovascular disease, whether framed as nutrient composition, dietary patterns, or broader food-system measures.[20-23] Within NHANES, dietary analyses have linked ultraprocessed foods, inflammatory indices, nutrient-wide profiles, and broader dietary summaries to cardiovascular phenotypes or mortality, but those studies do not resolve the blockwise measurement question addressed here.[24-28] NHANES has also supported broader cardiovascular-health profiling within the same surveillance system.[29] A small residual increment from the dietary block would therefore speak to the marginal value of this dietary representation in this information sequence, not to the biologic importance of diet.

This was neither an incident-risk tool nor a causal analysis. Using pooled NHANES 2007–2018 data, we quantified the discrimination added by recall-derived dietary variables beyond demographic, lifestyle, and clinical blocks; compared logistic regression with histogram-based gradient boosting; and assessed whether the same hierarchy held in weighted, temporal, subgroup, block-order, and

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repeated stochastic imputation sensitivity analyses. The contribution is not the expected observation that clinical markers are strong classifiers of prevalent cardiovascular disease; that is well established. Rather, the contribution is the explicit blockwise quantification of residual information carried by a short-term dietary recall representation after demographic, lifestyle, and clinical information are already available. Prior NHANES machine-learning studies combined all available features and reported overall performance or variable rankings; they did not isolate blockwise residual increments through nested transitions, paired out-of-fold DeLong comparisons, held-out joint block permutation, block-order reversal, weighted sensitivities, subgroup-specific contrasts, and repeated stochastic imputation.[9,10,12] This framing is closer to recent Practice of Epidemiology work that uses machine learning to address methodological questions rather than to pursue performance gains alone.[30-34] We aligned the reporting structure to current guidance for observational and prediction-model studies.[35-38]

Methods

Study design, data source, and population

We used six pooled NHANES cycles from 2007 through 2018. NHANES is a nationally representative cross-sectional survey of the noninstitutionalized US population based on a complex, multistage sampling design.[39] Cycles 2007–2008 through 2017–2018 were selected because they provided a consistent, harmonizable set of the four self-reported cardiovascular outcome items, demographic and lifestyle questionnaire variables, Mobile Examination Center clinical and laboratory measures, and 24-hour dietary recall variables required for the prespecified Model A–C blocks across all six survey cycles. The analysis was restricted to this completed six-cycle period to preserve a common analytic specification across outcome, predictor, and dietary-recall components. All participants provide written informed consent, and the survey is conducted under National Center for Health Statistics ethics oversight.

The analytic file was restricted to adults aged 20 years or older. Participants with missing primary outcome status were excluded. Explicit current pregnancy, when indicated in the NHANES demographic file, was excluded before the final analytic set was constructed. The resulting primary analytic file contained 34 388 participants, of whom 3 492 (10.2%) were positive for the primary outcome. A common complete-case subset of 24 483 participants was retained for sensitivity analysis. Analytic-file flow and variable-level missingness are shown in Table S1.

Outcome, predictor blocks, and missing data

The primary outcome was a self-reported cardiovascular disease composite constructed from four NHANES Medical Conditions Questionnaire items: coronary heart disease (MCQ160C), angina or angina pectoris (MCQ160D), heart attack or myocardial infarction (MCQ160E), and stroke (MCQ160F). For each item, a response of 1 (yes) was coded as positive and 2 (no) as negative; responses coded 7 (refused) or 9 (don't know) were treated as missing. The composite was coded positive if any nonmissing component was reported as yes and negative if all nonmissing components were reported as no. Participants whose primary composite status was entirely missing were excluded before the analytic file was constructed. We use "self-reported cardiovascular disease composite" deliberately because the stroke item (MCQ160F) is not specific to atherosclerotic stroke. Restricted outcomes evaluated in sensitivity analyses were stroke excluded (MCQ160C, MCQ160D, MCQ160E), angina excluded (MCQ160C, MCQ160E, MCQ160F), and coronary heart disease or myocardial infarction only (MCQ160C, MCQ160E).

Predictors were organized into conceptually ordered measurement blocks. Model A contained demography and lifestyle: age, sex, race/ethnicity, education, poverty-income ratio, smoking status, vigorous physical activity, moderate physical activity, and sedentary time. Model B added body mass index, waist circumference, mean systolic blood pressure, mean diastolic blood pressure, hypertension history, diabetes status, total cholesterol, HDL cholesterol, and glycated hemoglobin A1c. Model C then added recall-derived dietary variables: total energy intake, macronutrient energy percentages, dietary fiber, added sugar, sodium, potassium, and recall-day count. These nutrient variables were derived from the NHANES 24-hour dietary recall module. Participants could contribute one or two completed recall days; recall-day count was retained so the models could account for whether the nutrient summary was based on 1 or 2 completed recalls. The dietary representation follows standard nutritional epidemiology practice by retaining energy-adjusted composition variables while treating short-term recall information as a limited proxy for longer-term intake.[17-19] The term "recall-derived" is used because the nutrient summary may incorporate either one or two completed 24-hour recall days.

This ordering represented information availability rather than causal order. The analytic question was whether recall-derived dietary variables added residual classification value after demographic, lifestyle, and downstream clinical markers were already observed.

The primary analysis used fold-specific median imputation for continuous variables and mode imputation for categorical variables because the goal was out-of-fold prediction rather than parameter estimation. To test whether the small B-to-C increment depended on that choice, we added a repeated stochastic imputation sensitivity. We repeated the full 5-fold validation procedure 5 times. Within each training fold, numeric variables were imputed with stochastic iterative imputation and categorical variables with the most frequent observed category; the fitted imputers were then applied to the held-out fold. Transition metrics were summarized across repetitions as mean and standard deviation. This analysis was used to assess robustness of prediction performance, not to implement a pooled inferential multiple- imputation framework.

Modeling, validation, and calibration

We evaluated logistic regression with L2 regularization and histogram-based gradient boosting. Logistic regression used standardized continuous variables and one-hot encoded categorical variables. For gradient boosting, categorical predictors were passed as native categorical features using pandas categorical dtype, rather than ordinal-threshold encoding. The modeling recipe was held fixed across Models A–C so that block-transition contrasts reflected information changes rather than step-specific retuning.

Two complementary model families were selected to assess whether blockwise contrasts were robust to a regularized linear model and a flexible nonlinear model, rather than to conduct an exhaustive algorithm comparison. L2-regularized logistic regression served as an interpretable conventional baseline. Histogram-based

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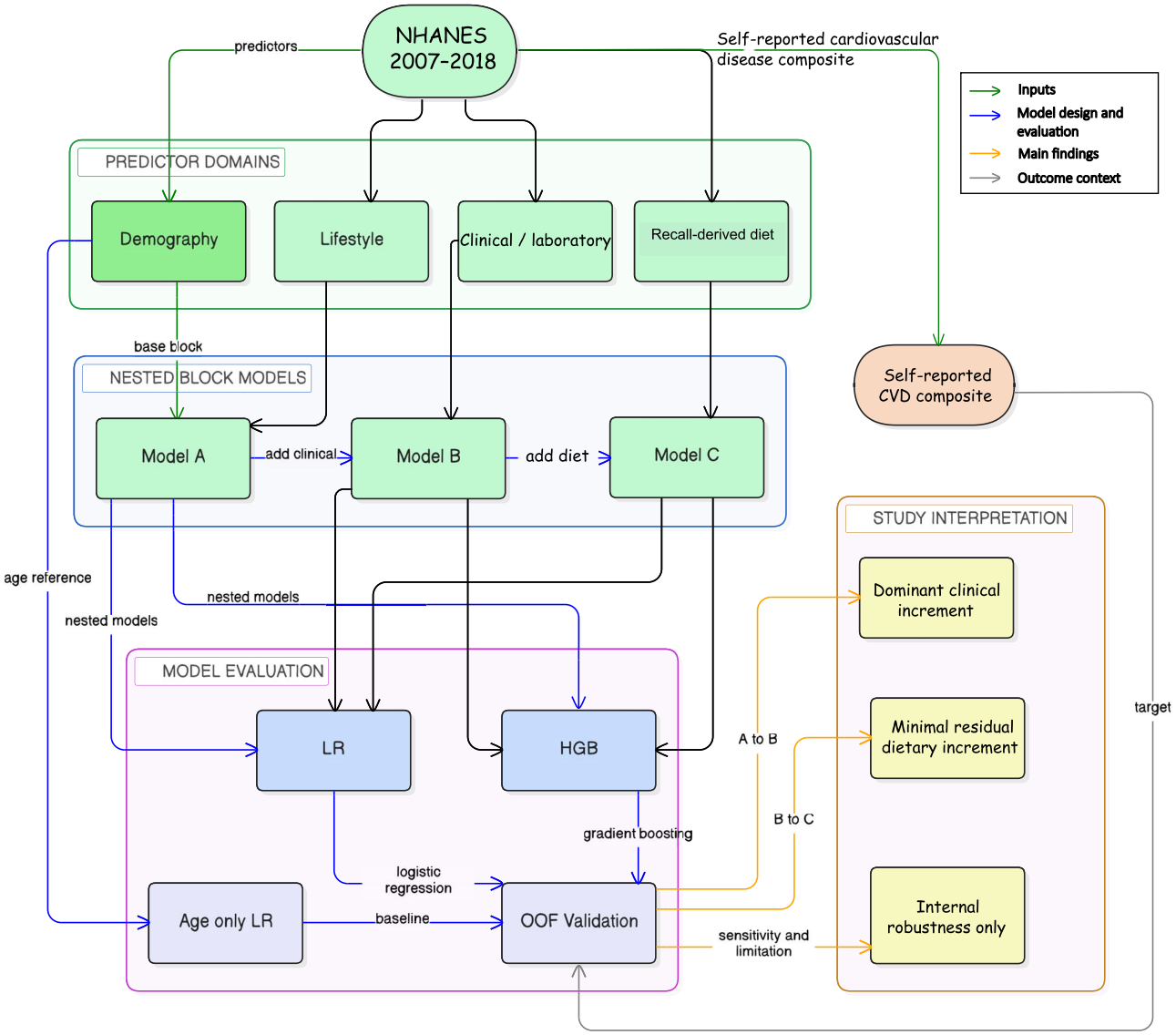


Figure 1 Study design and analytic framework for pooled National Health and Nutrition Examination Survey 2007–2018 data (United States; $n = 34{,}388$). Predictors were grouped into four domains and evaluated in three nested models: Model A (demography and lifestyle variables), Model B (Model A plus clinical and laboratory variables), and Model C (Model B plus recall-derived dietary variables). Five-fold stratified out-of-fold validation was used throughout. HGB, histogram-based gradient boosting; LR, logistic regression; ROC-AUC, area under the receiver operating characteristic curve.

gradient boosting served as a nonlinear comparator capable of capturing interactions and threshold effects without manual feature engineering. Additional algorithms, such as random forests or elastic-net logistic regression, were not included because the study objective was to estimate blockwise incremental value under fixed modeling recipes rather than to optimize performance across a large algorithm set.

The primary validation framework was 5-fold stratified out-of-fold validation. Discrimination was summarized with the area under the receiver operating characteristic curve (ROC-AUC) and precision-recall area under the curve (PR-AUC). Overall probabilistic prediction error was summarized with the Brier score. Calibration was summarized numerically using calibration slope, calibration intercept, observed-to-expected ratio, expected calibration error, and Brier skill score, and visually with calibration curves for Models B and C.[40,41] We did not prioritize reclassification-style indices because the central estimand was incremental discrimination after earlier information blocks were already observed.[42] We judged incremental value by effect size and consistency across model families, calibration summaries, and robustness analyses rather than by nominal P values alone. A transition was treated as materially informative only when the direction and magnitude of the gain persisted across these checks; otherwise very small deltas were interpreted as practically minimal in this setting.

Paired DeLong comparisons were applied to pooled out-of-fold predictions for the A-to-B and B-to-C ROC-AUC transitions and are interpreted as approximate because the underlying models were refit across folds. Predictor-domain contribution was summarized as held-out joint block permutation loss in ROC-AUC for Model C. For subgroup analyses, we additionally summarized subgroup-specific Model B and Model C ROC-AUC values, subgroup-specific B-to-C ΔROC-AUC, bootstrap percentile confidence intervals for that increment based on

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Table 1 Weighted baseline characteristics of adults included in the primary analytic sample by self-reported cardiovascular disease composite status, United States, NHANES 2007–2018.

Characteristic	Composite-negative	Composite-positive	SMD
Demography / socioeconomic position			
Age, y	46.2 (16.5)	65.0 (12.9)	1.272
Female	52.0	44.4	0.152
Mexican American	8.9	4.7	0.168
Other Hispanic	6.1	4.0	0.097
Non-Hispanic White	65.5	72.9	0.161
Non-Hispanic Black	11.4	11.1	0.009
Other / multiracial	8.2	7.4	0.031
Education <9th grade	5.2	9.3	0.159
Education 9–11th grade	10.2	14.1	0.117
High school / GED	22.6	27.9	0.122
Some college / AA	31.4	28.3	0.070
College graduate	30.4	20.3	0.232
Poverty-income ratio	3.0 (1.6)	2.6 (1.6)	0.272
Lifestyle / behavior			
Current smoker	19.6	22.0	0.058
Vigorous PA	22.1	15.9	0.158
Moderate PA	41.5	34.1	0.153
Sedentary time, min/day	369.5 (202.9)	400.4 (209.5)	0.150
Clinical / laboratory			
BMI, kg/m^2	29.0 (6.9)	30.4 (7.0)	0.205
Waist circumference, cm	98.7 (16.5)	106.1 (16.0)	0.456
SBP, mmHg	121.9 (16.9)	129.8 (20.6)	0.417
DBP, mmHg	71.2 (11.5)	68.1 (13.5)	0.250
Hypertension history	28.9	71.9	0.953
Diabetes	8.1	31.4	0.611
Total cholesterol, mg/dL	194.2 (40.8)	179.3 (44.3)	0.350
HDL cholesterol, mg/dL	53.6 (16.4)	50.1 (16.5)	0.209
HbA1c, %	5.6 (0.9)	6.2 (1.2)	0.526
Recall-derived dietary representation			
Energy intake, kcal/day	2120.1 (860.6)	1892.9 (775.6)	0.277
Protein, % energy	16.0 (4.5)	15.9 (4.3)	0.035
Carbohydrate, % energy	47.9 (9.8)	48.5 (10.0)	0.068
Total fat, % energy	34.4 (7.8)	35.0 (7.9)	0.076
Saturated fat, % energy	11.2 (3.3)	11.4 (3.4)	0.067
MUFA, % energy	12.2 (3.3)	12.4 (3.3)	0.060
PUFA, % energy	7.9 (2.9)	8.0 (3.2)	0.035
Dietary fiber, g/1000 kcal	8.4 (4.0)	8.6 (3.8)	0.035
Added sugar, g/1000 kcal	52.1 (22.4)	53.4 (23.7)	0.054
Sodium, mg/1000 kcal	1694.3 (744.7)	1713.7 (462.4)	0.031
Potassium, mg/1000 kcal	1316.7 (411.2)	1400.4 (434.4)	0.198
Dietary recall days	1.8 (0.6)	1.8 (0.5)	0.033

Abbreviations: BMI, body mass index; DBP, diastolic blood pressure; GED, General Educational Development; HbA1c, glycated hemoglobin A1c; HDL, high-density lipoprotein; MUFA, monounsaturated fatty acids; NHANES, National Health and Nutrition Examination Survey; PA, physical activity; PUFA, polyunsaturated fatty acids; SBP, systolic blood pressure; SMD, standardized mean difference.*n* = 34,388 (composite-negative: 30,896; composite-positive: 3,492). Continuous values are weighted mean (SD); categorical values are weighted %. Pooled MEC sample weights were applied for all weighted estimates.

subgroup-restricted out-of-fold predictions, and calibration slopes for both models.

Sensitivity analyses and survey weights

Sensitivity analyses included common complete-case analysis, restricted outcomes, component-specific individual outcome analyses, temporal validation (train cycles E–H; test cycles I–J), recall-day removal, parsimonious nutrition specifications, weighted performance summaries, weighted-fit logistic sensitivity, subgroup performance summaries, subgroup-specific incremental contrasts, block-order reversal, and age-dominance checks. The block-order analysis compared the standard sequence of demography/lifestyle, then clinical, then nutrition variables with a reversed sequence in which nutrition preceded the clinical block. Outcome, temporal, and block-order sensitivity results are shown in Supplementary Table S2. Component-specific individual outcome analyses are reported in the Supplementary Data.

Descriptive statistics in Table 1 use pooled MEC weights. Primary model fitting remained unweighted because the estimand was discrimination in the observed analytic sample, not nationally representative coefficient estimation. To address survey-design concerns directly, we added two weight-related sensitivities. First, we recalculated discrimination and calibration by applying pooled MEC weights to the primary out-of-fold predictions; this was a weighted performance-summary sensitivity, not weighted model fitting. Second,

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Table 2 Primary discrimination and Brier performance from 5-fold out-of-fold predictions, imputed full sample and complete-case subset, United States, NHANES 2007–2018.

Model	ROC-AUC (95% CI)	PR-AUC (95% CI)	Brier (95% CI)
Imputed full sample (n = 34,388)			
LR age-only	0.795 (0.788–0.802)	0.262 (0.251–0.274)	0.082 (0.079–0.084)
LR Model A	0.819 (0.812–0.825)	0.306 (0.292–0.320)	0.079 (0.077–0.081)
LR Model B	0.849 (0.843–0.855)	0.373 (0.358–0.388)	0.075 (0.073–0.077)
LR Model C	0.851 (0.845–0.857)	0.374 (0.360–0.389)	0.075 (0.073–0.077)
HGB Model A	0.821 (0.815–0.828)	0.317 (0.302–0.331)	0.079 (0.076–0.081)
HGB Model B	0.853 (0.847–0.859)	0.377 (0.362–0.394)	0.074 (0.072–0.076)
HGB Model C	0.854 (0.848–0.860)	0.376 (0.362–0.393)	0.074 (0.072–0.076)
Complete-case subset (n = 24,483)			
LR age-only	0.793 (0.785–0.802)	0.255 (0.241–0.269)	0.078 (0.075–0.080)
LR Model A	0.819 (0.811–0.826)	0.302 (0.286–0.319)	0.075 (0.073–0.078)
LR Model B	0.850 (0.844–0.857)	0.363 (0.345–0.383)	0.071 (0.069–0.074)
LR Model C	0.852 (0.845–0.859)	0.363 (0.343–0.383)	0.071 (0.069–0.074)
HGB Model A	0.820 (0.813–0.827)	0.308 (0.292–0.325)	0.075 (0.073–0.077)
HGB Model B	0.853 (0.847–0.861)	0.372 (0.353–0.391)	0.071 (0.068–0.073)
HGB Model C	0.856 (0.849–0.863)	0.373 (0.352–0.393)	0.070 (0.068–0.073)

Abbreviations: Brier, Brier score; CI, confidence interval; HGB, histogram-based gradient boosting; LR, logistic regression; NHANES, National Health and Nutrition Examination Survey; PR-AUC, area under the precision-recall curve; ROC-AUC, area under the receiver operating characteristic curve. Model A: demography and lifestyle variables. Model B: Model A plus clinical and laboratory variables. Model C: Model B plus recall-derived dietary variables. 95% CIs from 5-fold stratified out-of-fold predictions.

for logistic regression only, we fit Models A–C within each training fold using pooled MEC weights as sample weights and then evaluated weighted out-of-fold performance. These sensitivities are reported in Table S3.

Primary model-performance estimates are unweighted and describe discrimination in the observed analytic sample. They should not be interpreted as design-based nationally representative model-performance estimates. Survey weights were used for descriptive statistics (Table 1) and in weighted sensitivity analyses (Table S3) to assess robustness. Neither the weighted performance summaries nor the weighted-fit logistic sensitivity constitutes full complex-survey modeling or population-level coefficient inference.

All analyses were performed in Python 3.12.

Ethics approval

NHANES is conducted under National Center for Health Statistics ethics oversight, and all participants provided written informed consent. This secondary analysis used de-identified public-use data; no additional institutional review board approval was required.

Results

Study population and baseline structure

The imputed full sample contained 34 388 participants, including 3 492 composite-positive observations. The common complete-case subset contained 24 483 participants and 2 344 composite-positive observations. Weighted baseline characteristics are shown in Table 1. Composite-positive participants were older and carried a higher clinical burden than composite-negative participants, with the largest standardized differences in age, hypertension history, glycated hemoglobin A1c, and waist circumference. Missingness was concentrated in poverty-income ratio and the nutrition variables (Table S1), which motivated the prediction- oriented imputation strategy. The variable-level missingness profile is also shown in Figure S1. Most short-term nutrient variables showed more limited separation, consistent with the expectation that a brief dietary snapshot carries less direct cross-sectional signal than age and downstream clinical markers.

Primary discrimination and calibration

Primary out-of-fold performance is summarized in Table 2 and Figure 2. The age-only logistic baseline reached ROC-AUC 0.795. Expanding to the full demography/lifestyle block increased ROC-AUC to 0.819 in logistic regression and 0.821 in gradient boosting. The dominant gain occurred when the clinical/laboratory block was added: A-to-B ΔROC-AUC was 0.031 (95% confidence interval: 0.027, 0.034; $P < .001$) in logistic regression and 0.032 (95% confidence interval: 0.028, 0.036; $P < .001$) in gradient boosting.

The dietary block added little after the clinical block. In logistic regression, the B-to-C ROC-AUC increment was 0.002 (95% confidence interval: 0.001, 0.003; $P < .001$). In gradient boosting, the corresponding increment was 0.001 (95% confidence interval: 0.000, 0.002; $P = .219$). PR-AUC and Brier changes at the B-to-C step were similarly small. Under the study's effect-size interpretation framework, the logistic-regression B-to-C increment was treated as statistically detectable but practically minimal because its magnitude remained very small and did not become materially larger in the robustness analyses.

Numeric calibration summaries are reported in Table S3, and calibration curves are shown in Figure 3. Calibration slopes for the primary Models B and C ranged from 0.989 to 1.049, observed-to-expected ratios stayed near 1.00, and expected calibration error remained small. The B-to-C transition did not materially worsen calibration in either model family.

Sensitivity analyses and block-order interpretation

Outcome restriction, temporal validation, and block-order results appear in Table S2. The main hierarchy remained unchanged across all restricted outcomes: the clinical block produced the large gain, whereas the dietary block remained small. In temporal validation,

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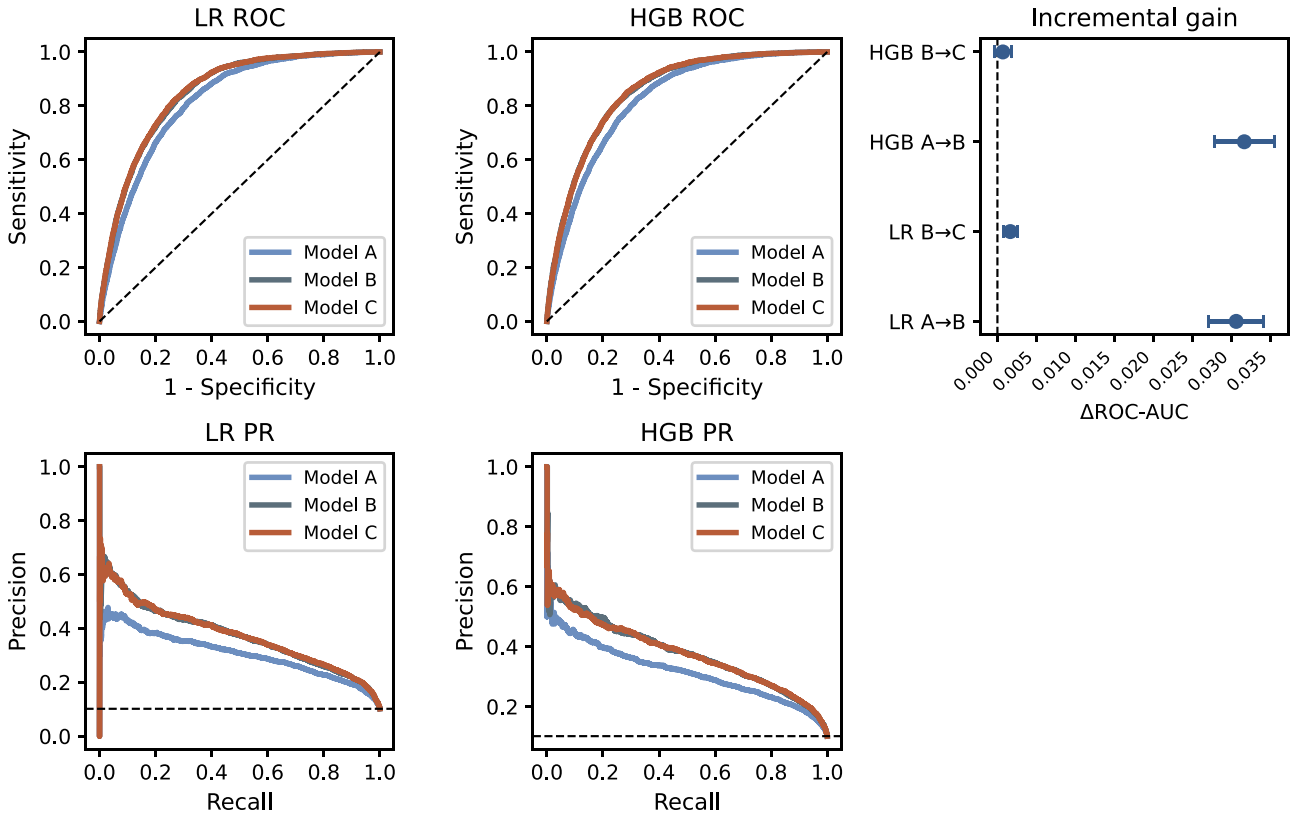


Figure 2 Primary discrimination curves and block-transition gains for the imputed full sample (United States, National Health and Nutrition Examination Survey 2007–2018; n = 34,388). A) Receiver operating characteristic curves for logistic regression Models A, B, and C. B) Receiver operating characteristic curves for histogram-based gradient boosting Models A, B, and C. C) Block-transition ΔROC-AUC point estimates and 95% confidence intervals from DeLong paired comparisons of pooled out-of-fold predictions. HGB, histogram-based gradient boosting; LR, logistic regression; ROC-AUC, area under the receiver operating characteristic curve. Model A included demography and lifestyle variables; Model B added clinical and laboratory variables; Model C added recall-derived dietary variables.

Model C ROC-AUC was 0.848 for logistic regression and 0.850 for gradient boosting on cycles I–J. Component-specific individual outcome analyses for coronary heart disease, angina, myocardial infarction, and stroke showed the same qualitative pattern. Across these individual components, the B-to-C increment was no larger than 0.004 for gradient boosting and 0.003 for logistic regression.

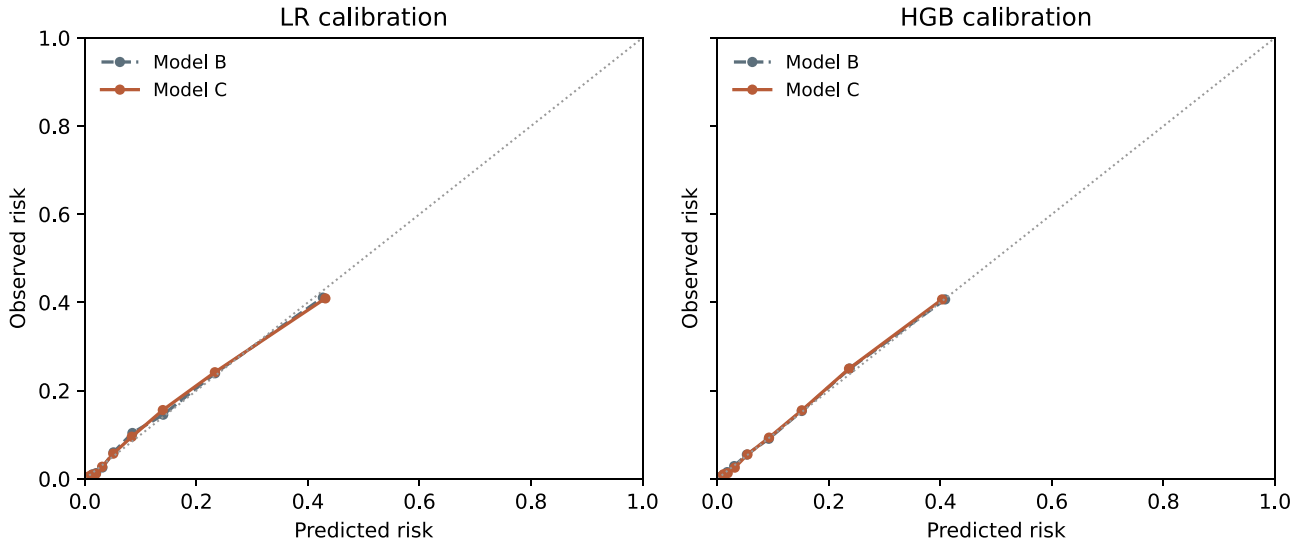


Figure 3 Calibration curves for Models B and C in logistic regression and histogram-based gradient boosting, pooled National Health and Nutrition Examination Survey 2007–2018, United States (n = 34,388). Panels show observed event fractions plotted against mean predicted probabilities in deciles of predicted risk. The diagonal line represents perfect calibration. HGB, histogram-based gradient boosting; LR, logistic regression. Model A included demography and lifestyle variables; Model B added clinical and laboratory variables; Model C added recall-derived dietary variables.

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Table 3 Block-transition discrimination gains and grouped held-out joint permutation importance, imputed full sample, United States, NHANES 2007–2018.

Panel A. Block-transition ΔROC-AUC

Algorithm	Transition	ΔROC-AUC	95% CI	*P*
LR	A→B	0.031	[0.027, 0.034]	<.001
LR	B→C	0.002	[0.001, 0.003]	<.001
HGB	A→B	0.032	[0.028, 0.036]	<.001
HGB	B→C	0.001	[−0.000, 0.002]	=.219

Panel B. Grouped held-out joint permutation loss in ROC-AUC (Model C)

Algorithm	Block	Total loss	Mean/feature	*n* features
LR	Demography	0.193	0.039	5
LR	Behavior	0.010	0.002	4
LR	Clinical	0.067	0.007	9
LR	Nutrition	0.004	0.000	12
HGB	Demography	0.208	0.042	5
HGB	Behavior	0.007	0.002	4
HGB	Clinical	0.081	0.009	9
HGB	Nutrition	0.003	0.000	12

Abbreviations: CI, confidence interval; HGB, histogram-based gradient boosting; LR, logistic regression; NHANES, National Health and Nutrition Examination Survey; ROC-AUC, area under the receiver operating characteristic curve. Model A: demography and lifestyle variables. Model B: Model A plus clinical and laboratory variables. Model C: Model B plus recall-derived dietary variables. Transition *P* values from DeLong paired comparisons on pooled out-of-fold predictions; these are approximate because models were refit across folds. Panel B reports mean ROC-AUC loss after held-out joint permutation of all features in each block simultaneously within Model C.

The block-order sensitivity is central to interpretation. When nutrition was added before the clinical block, the A-to-nutrition increment remained small: 0.003 in logistic regression and 0.002 in gradient boosting. After nutrition had already been added, the subsequent clinical block still increased ROC-AUC by 0.029 and 0.031, respectively.

Weight-related sensitivity analyses also preserved the same pattern (Table S3). In the weighted-fit logistic sensitivity, the A-to-B increment was 0.028 (95% confidence interval: 0.023, 0.033) and the B-to-C increment was 0.001 (95% confidence interval: −0.0005, 0.0024). Prediction-oriented repeated stochastic imputation sensitivity also preserved the same pattern (Table S4); across 5 repetitions, the mean B-to-C increment was 0.0015 for logistic regression and 0.0004 for histogram-based gradient boosting.

Subgroup performance and predictor-domain contribution

Grouped held-out joint permutation results are shown in Table 3. In both model families, the dominant contribution came from demography and the clinical block, with only a small joint contribution from the nutrition block.

Figure 4 summarizes the main robustness pattern across complete-case, restricted-outcome, temporal, recall-day, and parsimonious-nutrition checks. Table S5 reports subgroup-specific Model B-to-Model C contrasts, and Table S4 summarizes Model C subgroup calibration and representation-focused sensitivities. Across examined strata, subgroup-specific B-to-C contrasts remained small: logistic-regression deltas ranged from −0.002 to 0.007, and gradient-boosting deltas ranged from −0.010 to 0.003. No subgroup showed a materially favorable reversal of the overall hierarchy. Model C ROC-AUC ranged from 0.711 to 0.895 and calibration slopes ranged from 0.766 to 1.238 across subgroups.

The nutrient representation also remained structurally correlated. In the full nutrition block, the largest variance inflation factors were 131.3 for total fat (% energy), 30.3 for saturated fat, 27.9 for monounsaturated fatty acids, and 20.1 for polyunsaturated fatty acids; simplification reduced but did not eliminate this dependence, with the largest simplified-block factor remaining 5.7 for carbohydrate (% energy) (Table S4). These diagnostics describe the compositional correlation within the nutrition representation; the primary block-level interpretation does not rely on individual nutrient coefficient estimates, but on out-of-fold contrasts between nested predictor sets. Component-specific individual outcome results are provided in Table S6.

Discussion

In pooled NHANES 2007–2018 data, most discrimination for the self-reported cardiovascular disease composite came from age and the clinical/laboratory block. After those variables were included, the short-term dietary block added little: a small increment in logistic regression and no material increment in histogram-based gradient boosting. The block-order sensitivity supported the same interpretation. When nutrition was added before the clinical block, its gain remained small and the subsequent clinical block still produced the larger increase. In this setting, the logistic-regression B-to-C increment is best viewed as statistically detectable but practically minimal.

That pattern is consistent with the limited resolution of one- or two-day dietary recalls and with the fact that dietary variables were evaluated after demographic, lifestyle, and contemporaneously measured clinical information was available. The resulting contrasts quantify residual cross-sectional classification information for this measurement sequence. They do not estimate etiologic effects of diet on cardiovascular disease or its biological importance.

The main contribution is methodological. The finding that clinical markers are strong classifiers of prevalent cardiovascular disease is expected and well established. What has been less clear is the explicit magnitude and robustness of the residual increment from

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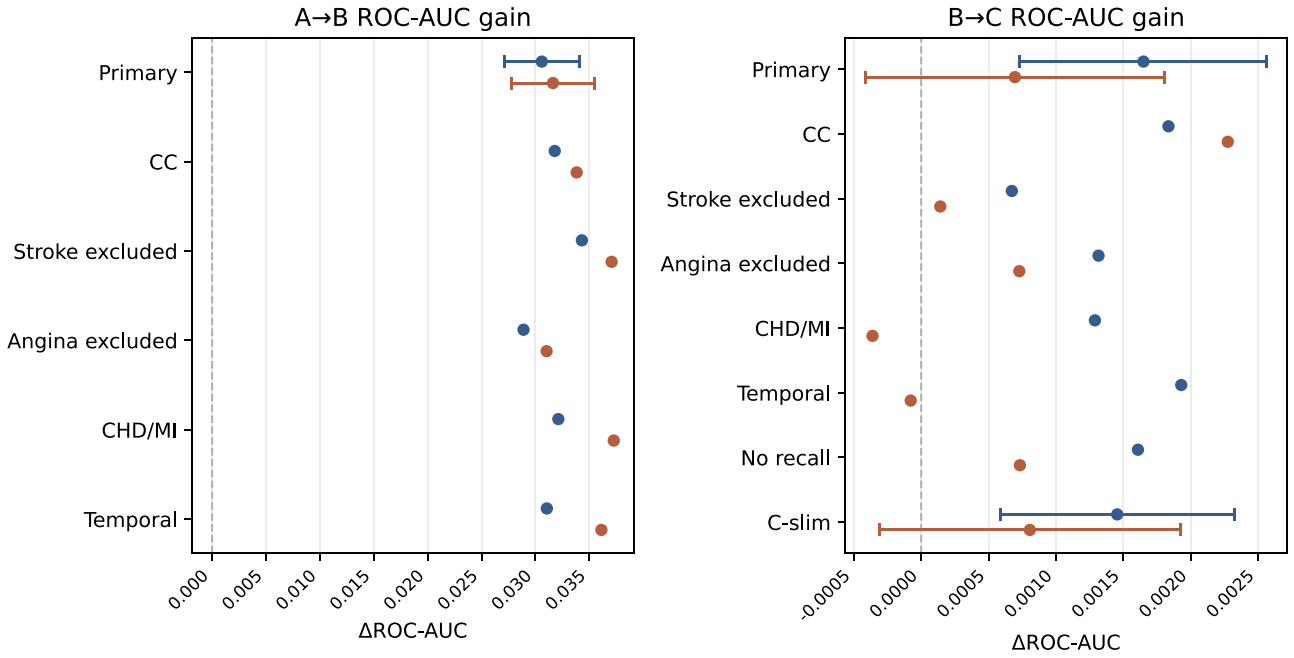


Figure 4 Robustness of A-to-B and B-to-C ROC-AUC gains across sensitivity analyses, National Health and Nutrition Examination Survey 2007–2018, United States. Each row represents one sensitivity analysis. Point estimates and 95% confidence intervals are shown for logistic regression (LR) and histogram-based gradient boosting (HGB). CHD, coronary heart disease; HGB, histogram-based gradient boosting; LR, logistic regression; MI, myocardial infarction; ROC-AUC, area under the receiver operating characteristic curve. Model A included demography and lifestyle variables; Model B added clinical and laboratory variables; Model C added recall-derived dietary variables.

short-term dietary recall data after those clinical markers are already available. The nested-block design, paired transition testing, held-out joint block permutation, block-order reversal, weighted sensitivities, subgroup-specific B-to-C contrasts, and repeated stochastic imputation together provide a fuller view of the measurement hierarchy than a single benchmark table. This approach complements prior NHANES-based machine-learning studies that emphasized overall predictive performance or post hoc importance rankings rather than explicit blockwise incremental value.[9,10,12] It is also distinct from recent Practice of Epidemiology applications that used machine learning to address different methodological questions, including neighborhood overdose prediction, model transport, phenotype classification, explainable mortality prediction, and nationwide registry screening for suicide-attempt risk.[30-34]

Subgroup analyses did not materially change that interpretation. B-to-C contrasts remained small across examined strata, and no subgroup showed a favorable reversal of the overall hierarchy. Discrimination and calibration varied across strata, as expected in cross-sectional survey data with different prevalence and case-mix structures. The survey-design sensitivities likewise preserved the same practical pattern. These weighted analyses should be read as robustness checks rather than as full complex-survey modeling. The primary performance estimates are analytic-sample quantities and do not establish design-based national transportability or complex-survey coefficient inference.

Several limitations remain. The outcome is self-reported and clinically heterogeneous by construction. Component-specific analyses reduced but did not remove this concern because the individual outcomes were also self-reported and had different event counts. The cross-sectional design also allows the observed predictor pattern to reflect medication use, physician counseling, diagnosis-related dietary modification, and accumulated disease burden. The dietary block is a correlated nutrient-density representation with residual compositional dependence. Temporal validation within NHANES is not external validation. More generally, studies that combine regression and machine- learning models still require careful reporting and applicability assessment.[35-37,43]

Within pooled US cross-sectional survey data, demographic and especially clinical information accounted for most of the discrimination for the self-reported cardiovascular disease composite. Once those variables were observed, the short-term dietary block added little. The main contribution of this study is therefore a blockwise account of measurement hierarchy in this setting, not an etiologic claim about diet.

Supplementary material

Supplementary material is available at *American Journal of Epidemiology* online.

Conflicts of interest

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Data availability

The data underlying this article are publicly available from the National Center for Health Statistics through the National Health and Nutrition Examination Survey (NHANES) website (https://wwwn.cdc.gov/nchs/nhanes/Default.aspx). Analytic code supporting the findings of this study is available from the corresponding author upon reasonable request.

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